

Financial Econometrics - Exercise 3.31

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Exercise 3.31: Demand Analysis for Canned Tuna

This study analyzes 52 weeks of sales data for a major brand of canned tuna. We explore the relationship between unit sales (SAL1) and the price per can (APR1).

Part (a): Summary Statistics and Time Series Analysis

Answer:

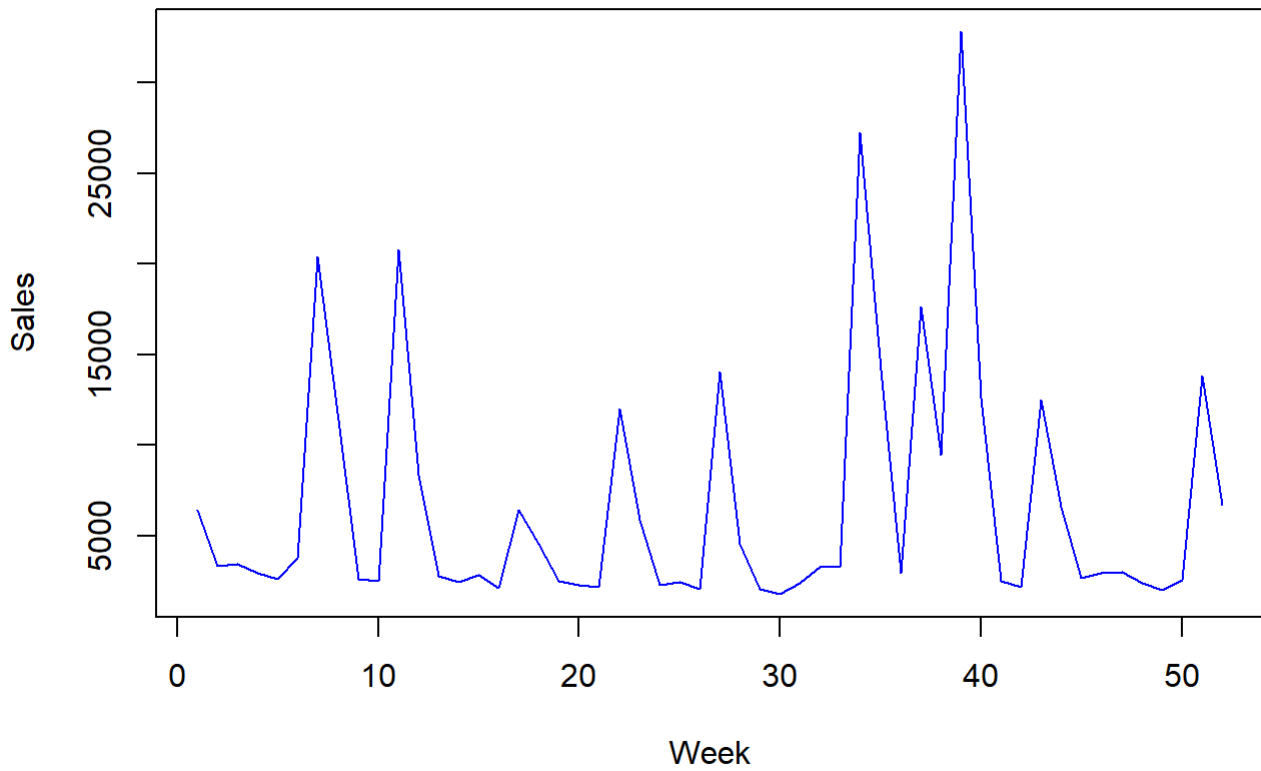
- **Summary Statistics:** The sample mean for weekly sales is 6718.7 units with a very high standard deviation of 6915.1, indicating significant volatility. In contrast, the price per can (APR1) is relatively stable, averaging \$0.78 with a standard deviation of only \$0.098.
- **Time Series Observation:** Plotting these variables against time (WEEK) reveals that sales spikes often coincide with price drops. This suggests that consumers are responsive to promotional pricing throughout the year.

```
# --- (a) Summary Statistics & Time Series Plots ---  
# 計算 SAL1 與 APR1 的平均、最小、最大、標準差  
  
summary_stats <- data.frame(  
  Variable = c("SAL1", "APR1"),  
  Mean = c(mean(tuna$sal1), mean(tuna$apr1)),  
  Min = c(min(tuna$sal1), min(tuna$apr1)),  
  Max = c(max(tuna$sal1), max(tuna$apr1)),  
  SD = c(sd(tuna$sal1), sd(tuna$apr1))  
)  
print(summary_stats)
```

##	Variable	Mean	Min	Max	SD
## 1	SAL1	6718.7115	1762.00	32820.00	6.915083e+03
## 2	APR1	0.7825	0.61	0.92	9.803711e-02

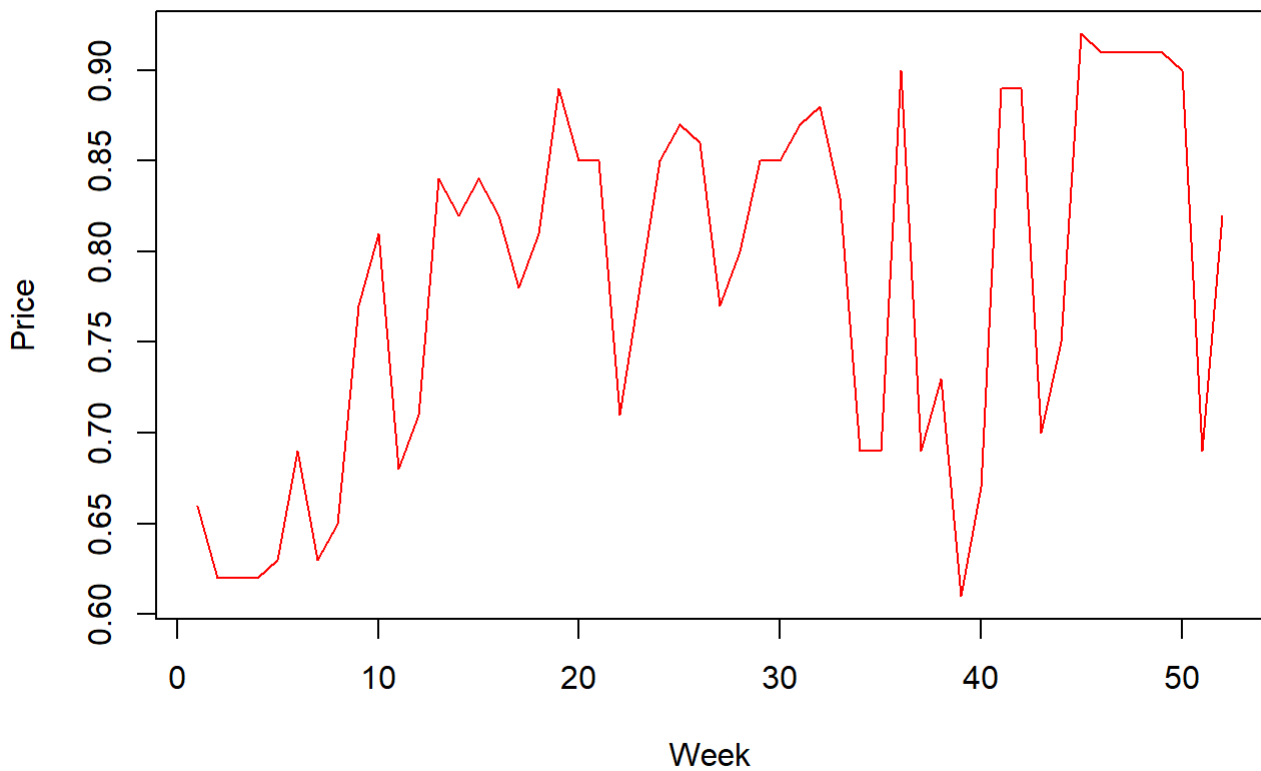
```
# 繪製隨時間變動的圖(時間序列圖)  
###取消 僅學習par(mfrow = c(2, 1)) # 分成上下兩格圖###  
plot(1:52, tuna$sal1, type = "l", col = "blue", main = "Weekly Sales (SAL1)"  
     , xlab="Week", ylab="Sales")
```

Weekly Sales (SAL1)



```
plot(1:52, tuna$apr1, type = "l", col = "red", main = "Weekly Price (APR1)"  
     , xlab="Week", ylab="Price")
```

Weekly Price (APR1)



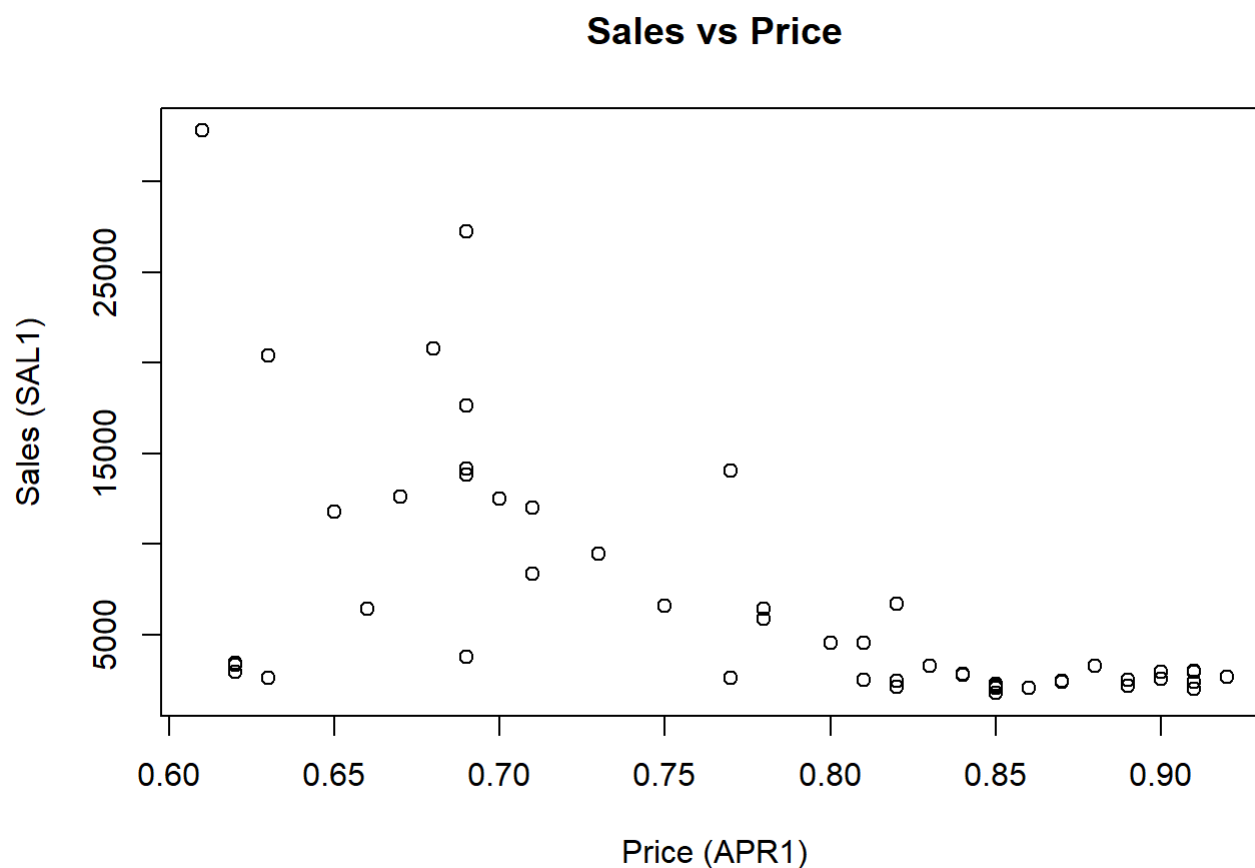
```
###取消 僅學習par(mfrow = c(1, 1)) # 恢復單圖模式###
```

Part (b): Sales vs. Price Relationship

Answer:

- **Visual Relationship:** The scatter plot shows a clear inverse (negative) relationship between price and sales. As the price per can increases, the number of units sold tends to decrease.
- **Economic Expectation:** This result is exactly as expected under the Law of Demand, which states that, ceteris paribus, the quantity demanded of a good falls when the price of the good rises.

```
# --- (b) Scatter Plot 散佈圖 ---  
plot(tuna$apr1, tuna$sal1, xlab = "Price (APR1)", ylab = "Sales (SAL1)",  
      main = "Sales vs Price")
```



Part (c): Linear Regression and the Effect of Price (in Cents)

Answer:

- **Point Estimate:** The coefficient β_2 represents the estimated change in unit sales for every one-cent increase in the price. Based on our model, a one-cent increase in price leads to an estimated decrease of -434.4473 units sold.

- **95% Interval Estimate:** In repeated sampling, approximately 95% of the intervals constructed using this methodology will contain the true value of the effect of a one-cent price increase on sales. For this sample, the interval is $[-592.2897, -276.6049]$.

```
# --- (c) Linear Regression (Unit: Cents) ---
tuna$price1 <- 100 * tuna$apr1 # 換算成 cent
mod_tuna <- lm(sal1 ~ price1, data = tuna)
summary(mod_tuna)
```

```
##
## Call:
## lm(formula = sal1 ~ price1, data = tuna)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10869.5  -1588.3   -499.3   1626.2  18607.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  40714.22    6196.42   6.571 2.82e-08 ***
## price1       -434.45     78.58  -5.528 1.17e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5502 on 50 degrees of freedom
## Multiple R-squared:  0.3794, Adjusted R-squared:  0.367
## F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

```
# 提取 Beta2 估計值與 95% 信賴區間
b2_tuna <- coef(mod_tuna)["price1"]
ci_b2 <- confint(mod_tuna, "price1", level = 0.95)

cat("Point Estimate (Beta2): ", b2_tuna, "\n")
```

```
## Point Estimate (Beta2):  -434.4473
```

```
cat("95% CI for Beta2: [", ci_b2[1], ",", ci_b2[2], "]\n")
```

```
## 95% CI for Beta2: [ -592.2897 , -276.6049 ]
```

Part (d): Expected Sales at 70 Cents

Answer:

- **Prediction:** For a week where the price is set at 70 cents ($PRICE1 = 70$), the expected number of cans sold is $[10302.9]$.
- **90% Interval Estimate:** Using the principle of repeated sampling, we expect that 90% of such intervals calculated from different samples would capture the true population mean sales at this price point. Our calculated 90% interval is $[8624.934, 11980.87]$.

```
# --- (d) Expected Sales at 70 Cents ---
# 用 predict 函數計算 90% 信賴區間
new_data <- data.frame(price1 = 70)
predict_d <- predict(mod_tuna, new_data, interval = "confidence", level = 0.90)
print(predict_d)
```

```
##          fit          lwr          upr
## 1 10302.9 8624.934 11980.87
```

Part (e): Price Elasticity of Demand at the Means

Answer:

- **Elasticity Estimate:** The estimated price elasticity of demand, calculated at the sample means, is [-5.059825].
- **Interpretation:** This value implies that a 1% increase in price leads to an approximate [5.059825]% decrease in the quantity of tuna sold.
- **95% Interval Estimate:** In repeated sampling, approximately 95% of intervals constructed this way will contain the true price elasticity at the means. Our calculated 90% interval is [[-6.898149], [-3.221501]].
- **Economic Sense:** Since the absolute value of the elasticity is greater than 1, the demand for this brand of tuna is highly elastic. This makes economic sense because canned tuna is a non-essential commodity with many close substitutes (other brands or protein sources), making consumers very price-sensitive.

```
# --- (e) Elasticity at the Means ---
# Elasticity = Beta2 * (mean_p / mean_q)
mean_p <- mean(tuna$price1)
mean_q <- mean(tuna$sa11)
elas_mean <- b2_tuna * (mean_p / mean_q)

# 彈性的標準誤 (假設平均數為常數)
# SE(Elasticity) = SE(Beta2) * (mean_p / mean_q)
se_b2_tuna <- sqrt(vcov(mod_tuna)["price1", "price1"])
se_elas <- se_b2_tuna * (mean_p / mean_q)

# 95% 彈性信賴區間
df_tuna <- df.residual(mod_tuna)
t_crit_95 <- qt(0.975, df_tuna)
lower_elas <- elas_mean - t_crit_95 * se_elas
upper_elas <- elas_mean + t_crit_95 * se_elas

cat("Elasticity at the Means: ", elas_mean, "\n")
```

```
## Elasticity at the Means: -5.059825
```

```
cat("95% CI for Elasticity: [", lower_elas, ", ", upper_elas, "]\n")
```

```
## 95% CI for Elasticity: [ -6.898149 , -3.221501 ]
```

Part (f): Hypothesis Test for Elasticity

Answer:

- **Hypotheses:** We test the Null Hypothesis ($H_0 : \epsilon = -3$) against the Alternative Hypothesis ($H_1 : \epsilon \neq -3$).
- **Test Statistic and P-value:** The calculated t-statistic is [-2.250572] with a corresponding p-value of [0.02884204].
- **Conclusion:** Using a 10% significance level, we compare the p-value to $\alpha = 0.10$. If the p-value is less than 0.10, we reject the null hypothesis and conclude that the price elasticity is significantly different from -3. Otherwise, we fail to reject the null.

```
# --- (f) Hypothesis Test: Elasticity = -3 (10% Level) ---  
# H0: Elasticity = -3, H1: Elasticity != -3  
elas_null <- -3  
t_stat_f <- (elas_mean - elas_null) / se_elas  
p_val_f <- 2 * (1 - pt(abs(t_stat_f), df_tuna))  
  
cat("T-statistic for Hypothesis Test: ", t_stat_f, "\n")
```

```
## T-statistic for Hypothesis Test: -2.250572
```

```
cat("P-value: ", p_val_f, "\n")
```

```
## P-value: 0.02884204
```